

Style Transfer with Gesture Style Generator

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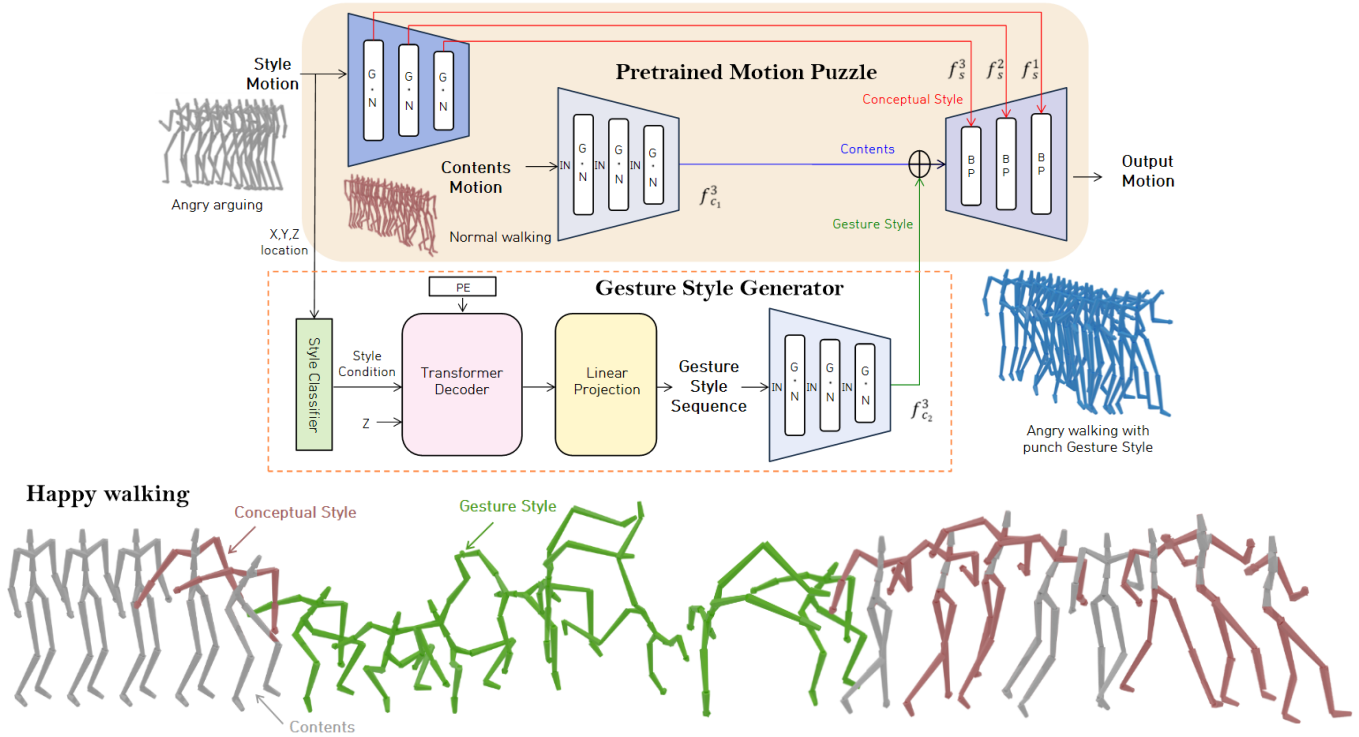


Figure 1: (a) Overall of our architecture (b) Motion sequence, which represent happily walking illustrates two distinct styles: Gesture Style, marked by the green-colored sections, and Conceptual Style, marked by the red-colored sections.

Abstract

Traditionally, style transfer methods have extracted style element from user-provided "Style motion" and content element from "Content motion", and combining these two elements to achieve transferred style in the content motion. In the process, they attempted to separate movements into style and content elements in a binary way. However, ambiguity exists between the definitions of style and content. An action of crossing your arms when you are angry (style) while walking (content) is not clearly categorized as either style or content. In this work we define "Gesture Style" referring to a short-term action revealing style (not content) in a motion sequence. Crossing arms in the example falls in Gesture Style, as it

represents anger but is performed for short time. We propose a new style transfer method with Gesture Style Generator. It transfers style to the output motion in the conventional style transfer manner while also incorporating generated Gesture Style.

CCS Concepts

• Computing methodologies → Motion processing.

Keywords

Motion style transfer, Motion generation, Style, Content

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1 Introduction

In previous style transfer methods, style and content have an overlapped definition. Previous studies failed to recognize this intersection of style and content characteristics, by treating it as style or contents element varying from time to time. We propose a new method of style transfer by dividing movements into three elements: Conceptual Style, Gesture Style, and Contents. The style defined in previous studies is the performing manner (such as proud or strut body movement) of a specific action (such as walking) rather than an action itself. So we refer the previous style to “Conceptual Style”. And we introduce a new element, “Gesture Style”, that is a short-term action revealing style (not content) in a long-term content motion. Unlike continuously appearing Conceptual Style, Gesture Style action is performed momentarily. Additionally, Gesture Style is distinct from Conceptual style in that the action itself conveys the style. We have grouped Gesture style and Conceptual style together under the term “Inclusive Style”. Regarding the new definitions, we combine existing style transfer model and motion generation model building “Style Transfer with Gesture Style Generator”. The Motion Puzzle[Jang et al. 2022] is a model capable of performing style transfer for individual body parts. The Action-Conditioned Motion Generation model[Guo et al. 2020] generates various motion sequences representing a given action. We improve Motion Puzzle style transfer model[Jang et al. 2022] by adding a “Gesture Style Generator” network. This allows not only for the transfer of Conceptual Style, but also for the recognition of the user’s intended Inclusive Style from the input motion and the generation of Gesture Style that reflects the Inclusive Style.

2 Proposed method

We develop a model capable of achieving style transfer by adding Gesture Style element. We utilize a Motion Puzzle[Jang et al. 2022] network. To generate Gesture Style, which is distinct from Conceptual Style, we propose an “Gesture Style Generator” network consisting of a Style Classifier and Gesture Style generator. This network was developed by modifying Action-conditioned motion generation model[Guo et al. 2020]. When a Style motion containing the desired Inclusive Style and a content motion are provided as inputs, our Style Classifier is designed to identify the specific Inclusive Style to be transferred. The Style Classifier extracts conditions for the Gesture Style generator. The Gesture Style generator consists of a decoder from a pretrained conditional Transformer-based VAE model. The generated style conditions are input into the conditional VAE decoder along with the Gesture Style representation vector Z . Based on these style conditions, the decoder generates a sequence of Gesture Style. Gesture Style is then transformed through a content encoder into a form to be added to content elements extracted from the input Content motion. At decoder combined contents feature values are converted into a motion with the style elements applied via the Bpstyle-net.

3 Experiment

We used a pretrained model for the Motion Puzzle[Jang et al. 2022] component and only trained our Gesture Style generator network. The network has been trained on two classes: happy and angry. The Style Classifier determines what Inclusive style the input style

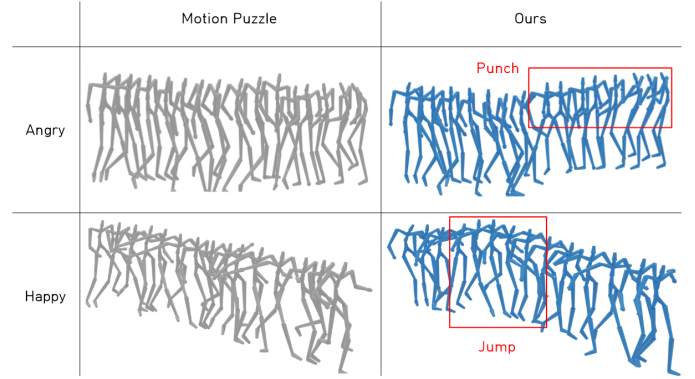


Figure 2: Comparison of Output Motions for Walking Contents Element between Two Classes

motion has based on the joint x, y, z positions and it is implemented with four fully connected layers. The Gesture Style Generator was pretrained with motions labeled with Inclusive style classes. Gesture Style motions such as kicking, punching were grouped into the “angry” class, while motions like laughing and tumbling were grouped into the “happy” class. We compared the results of our model with the Motion Puzzle[Jang et al. 2022] when using a Content motion representing walking as the input, combined with input Style motions representing the “angry” and “happy” Inclusive styles. As shown in Figure 2, in the case of the “Angry” class, our model successfully generated a punching Gesture Style, which aligns with the “angry” Inclusive style. For the “Happy” class, our model produced a jumping Gesture Style, reflecting the “happy” Inclusive style. By generating gesture styles, we were able to make the intended Inclusive style more pronounced in the output motion.

4 Conclusion and Limitation

We demonstrated the extensibility of the concept of style in style transfer through the incorporation of the Gesture Style element. Due to the lack of an optimal motion dataset representing Gesture Style, we trained our network using existing datasets to the fullest extent possible. By creating a dataset that effectively captures Gesture Style, we can expect even better results.

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